Personal Expense Tracker

Problem Statement:

Managing daily expenses is essential for financial planning. Many individuals struggle to keep track of their spending habits. This project aims to build a Personal Expense Tracker that allows users to record expenses, categorize them, and monitor their spending against a predefined monthly budget. The system also saves and loads expenses using file handling to ensure persistence.

Objectives:

- To develop a command-line based expense-tracking system
- To allow users to record daily expenses
- To categorize expenses such as Food, Travel, Shopping etc.
- To track total spending against a monthly budget
- To save and retrieve expense data using CSV files
- To provide an interactive menu-driven interface

Technologies Used:

Programming Language: Python

Concepts Used:

- Lists
- Dictionaries
- Functions
- Loops
- Conditional statements
- File Handling (CSV)
- Menu Driven Programming

System Design:

The system stores expenses in a list where each expense is represented as a dictionary containing:

date, category, amount, description

Example structure:

```
{
 'date': '2024-09-18',
 'category': 'Food',
 'amount': 150,
 'description': 'Lunch'
}
```

Features Implemented:

1. Add Expense
2. View Expenses
3. Set Monthly Budget
4. Track Budget
5. Save Expenses to CSV file
6. Load Expenses from CSV file
7. Menu-driven interface

Program Workflow:

Step 1: Program starts

Step 2: Load existing expenses from CSV file

Step 3: Display menu options

Step 4: User selects an option

Step 5: Corresponding function is executed

Step 6: Expenses are saved before exiting the program

Sample Output:

===== Personal Expense Tracker =====

1. Add Expense

2. View Expenses

3. Set Monthly Budget

4. Track Budget

5. Save Expenses

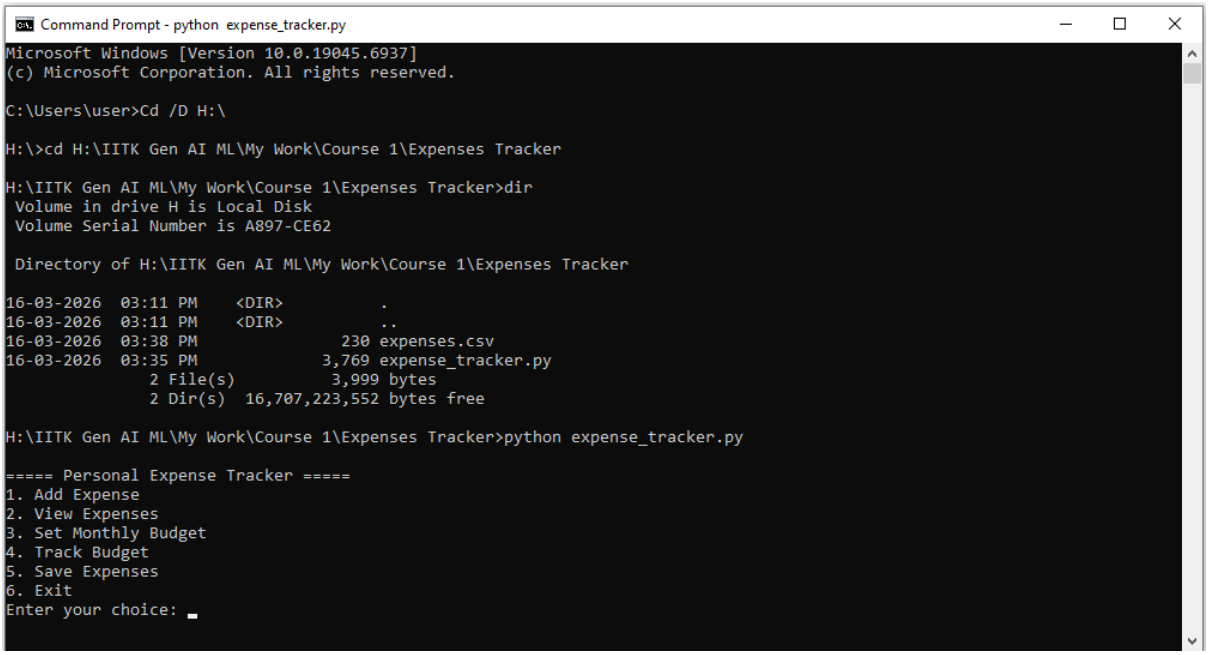
6. Exit

Conclusion:

The Personal Expense Tracker helps users maintain financial discipline by tracking daily expenses and monitoring spending against a monthly budget. The project demonstrates Python fundamentals such as data structures, file handling, and modular programming.

Screenshots:

SS 1: Running the Program & Menu Display



```
Command Prompt - python  expense_tracker.py
Microsoft Windows [Version 10.0.19045.6937]
(c) Microsoft Corporation. All rights reserved.

C:\Users\user>Cd /D H:\

H:\>cd H:\IITK Gen AI ML\My Work\Course 1\Expenses Tracker

H:\IITK Gen AI ML\My Work\Course 1\Expenses Tracker>dir
 Volume in drive H is Local Disk
 Volume Serial Number is A897-CE62

 Directory of H:\IITK Gen AI ML\My Work\Course 1\Expenses Tracker

16-03-2026  03:11 PM    <DIR>          .
16-03-2026  03:11 PM    <DIR>          ..
16-03-2026  03:38 PM               230 expenses.csv
16-03-2026  03:35 PM             3,769 expense_tracker.py
               2 File(s)          3,999 bytes
               2 Dir(s)  16,707,223,552 bytes free

H:\IITK Gen AI ML\My Work\Course 1\Expenses Tracker>python expense_tracker.py

===== Personal Expense Tracker =====
1. Add Expense
2. View Expenses
3. Set Monthly Budget
4. Track Budget
5. Save Expenses
6. Exit
Enter your choice:
```

SS 2: Adding an expense

```
Command Prompt - python  expense_tracker.py
16-03-2026  03:11 PM    <DIR>          .
16-03-2026  03:11 PM    <DIR>          ..
16-03-2026  03:38 PM               230 expenses.csv
16-03-2026  03:35 PM             3,769 expense_tracker.py
               2 File(s)          3,999 bytes
               2 Dir(s)  16,707,223,552 bytes free

H:\IITK Gen AI ML\My Work\Course 1\Expenses Tracker>python expense_tracker.py

===== Personal Expense Tracker =====
1. Add Expense
2. View Expenses
3. Set Monthly Budget
4. Track Budget
5. Save Expenses
6. Exit
Enter your choice: 1
Enter date (YYYY-MM-DD): 2025-11-06
Enter category (Food/Travel/etc): Travel
Enter amount spent: 2600
Enter description: Train to Chennai
Expense added successfully!

===== Personal Expense Tracker =====
1. Add Expense
2. View Expenses
3. Set Monthly Budget
4. Track Budget
5. Save Expenses
6. Exit
Enter your choice:
```

SS 3: Viewing Expenses

```
Command Prompt - python  expense_tracker.py

===== Personal Expense Tracker =====
1. Add Expense
2. View Expenses
3. Set Monthly Budget
4. Track Budget
5. Save Expenses
6. Exit
Enter your choice: 2

--- Expense List ---
Date: 2024-09-18
Category: Food
Amount: 15.5
Description: Lunch with friends
--------------------
Date: 2024-09-19
Category: Travel
Amount: 10.0
Description: Metro ticket
--------------------
Date: 2026-03-16
Category: Food
Amount: 1258.0
Description: Dinner Date
--------------------
Date: 2026-03-10
Category: Travel
Amount: 1650.0
Description: Trip to Chennai
--------------------
Date: 11-03-2026
Category: Food
Amount: 750.0
Description: Lunch in Chennai
--------------------
Date: 2025-11-06
Category: Travel
Amount: 2600.0
Description: Train to Chennai
--------------------

===== Personal Expense Tracker =====
```

SS 4: Set Budget

```
===== Personal Expense Tracker =====
1. Add Expense
2. View Expenses
3. Set Monthly Budget
4. Track Budget
5. Save Expenses
6. Exit
Enter your choice: 4
Please set a budget first.

===== Personal Expense Tracker =====
1. Add Expense
2. View Expenses
3. Set Monthly Budget
4. Track Budget
5. Save Expenses
6. Exit
Enter your choice: 3
Enter monthly budget: 4000
Budget set successfully!
```

SS 5: Budget Tracking

```
===== Personal Expense Tracker =====
1. Add Expense
2. View Expenses
3. Set Monthly Budget
4. Track Budget
5. Save Expenses
6. Exit
Enter your choice: 4

Total Expenses: 6283.5
Monthly Budget: 4000.0
⚠ You have exceeded your budget!
```